C 62588 23/9/19 (Pages : 4) Name.....

Reg. No.....

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION, MAY 2019

B.Sc.—Mathematics

MAT 2B 02—CALCULUS

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type)

Answer all twelve *questions.*
Each question carries 1 *mark.*

1. Find $\lim_{x \to -\infty} \frac{x^5}{x^6 + 2}$:

2. Find absolute extrema of $y = x^2$ on (0, 2).

3. Find dy if $y = 6 \cos x^5$.

4. Find the interval in which $y = x^3$ is concave up.

5. Suppose $\int_2^3 f(x)\,dx = -6$. Find $-\int_3^2 f(x)\,dx$.

6. Express the limit $\lim_{\|P\| \to 0} \sum_{k=1}^{n} \left(c_k^2 - 3c_k\right) \Delta x_k$ as an integral, where P is the partition of $[-7, 5]$.

7. Define average value of a function f on $[a, b]$.

8. Find all possible functions with derivative $y' = x^2$.

9. Evaluate $\int_0^{\pi/2} \sin^2 x dx$.

10. Write the sum $\sum_{k=1}^{5} k(3k + 5)$ without sigma notation.

11. State Mean Value Theorem.

12. Evaluate $\int_1^{10} x^{-2/3}\,dx$.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** *questions.*
Each question carries 2 marks.

13. Find $\lim_{t \to -\infty} \frac{2 - t + \sin t}{t + \cos t}$.

14. Evaluate $\sum_{k=1}^{9} k^3$.

15. Find the average value of the function $f(t) = \sin t$ on $[0, 2\pi]$.

16. Express the solution of the initial value problem $\frac{dy}{dx} = \tan x$, $y(1) = 5$ as an integral.

17. Find $\frac{dy}{dx} \int_0^{\sqrt{x}} \cos t\,dt$.

18. Find absolute extrema values of $g(t) = 8t - t^4$ on $[-2, 1]$.

19. Suppose that f is continuous and that $\int_0^5 f(z)\,dz = 3$ and $\int_0^6 f(z)\,dz = 7$. Find $\int_5^6 f(z)\,dz$.

20. The region between the curve $y = \sqrt{x}, 0 \le x \le 4$ and the x-axis is revolved about the x-axis to generate a solid. Find its volume.

21. Find the work done by the force $F(x) = \frac{1}{x^2}$ N along x-axis from $x = 1$ m to $x = 10$ m.

22. Find $\lim_{x \to \infty} \frac{\sin 2x}{x}$.

23. Evaluate $\int_0^{\pi/3} 2\sec^2 x dx$.

24. Find the interval in which $f(x) = -x^2 - 3x + 3$ is increasing and decreasing.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Find asymptotes of the graph of $f(x) = \frac{x^2 - 3}{2x - 4}$.

26. Find $\lim_{x \to \infty} \frac{2\sqrt{x} + x^{-1}}{3x - 7}$.

27. If b, c and d are constants, for what value of b will be the curve $y = x^3 + bx^2 + cx + d$ have a point of inflection at $x = 1$?

28. Suppose that $f(-1) = 3$ and that $f'(x) = 0$ for all $x \in R$. Must $f(x) = 3$? Give reasons for your answer.

29. Find the intervals on which $g(x) = -x^3 + 12x + 5, -3 \leq x \leq 3$ is increasing and decreasing. Where does the function assume extreme values and what are these values ?

30. Find the area of the region enclosed by $x = 2y^2$, $x = 0$ and $y = 3$.

31. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1$, $x = 4$ about the line $y = 1$.

32. Find the length of the curve $x = \frac{y^3}{3} + \frac{1}{4y}$ from $y = 1$ to $y = 3$.

33. Show that center of mass of a straight, thin strip or rod of constant density lies halfway between its two ends.

(6 × 5 = 30 marks)

Part D (Essay Type)

Answer any two questions.
Each question carries 10 marks.

34. Find the center of mass of a thin plate of constant density δ covering the legion bounded above by parabola $y = 4 - x^2$ and below by x-axis.

35. A spring has a natural length of 1 m. A force of 24 N stretches the spring to a length of 1.8 m.
    (a) Find the force constant k.
    (b) How much work will it take to stretch the spring 2 m. beyond its natural length ?
    (c) How far will a 45 N force stretch the spring ?

36. What values of a and b make $f(x) = x^3 + ax^2 + bx$ have,
    (a) A local maximum at $x = -1$ and a local minimum at $x = 3$.
    (b) A local minimum at $x = 4$ and a point of inflection at $x = 1$?

$(2 \times 10 = 20$ marks$)$